

Intelligent Document Assistant Sample

This sample document describes a production-style retrieval augmented generation system.

The application lets users upload PDF files, extracts text, cleans the content, and splits the text into overlapping chunks. Each chunk is embedded and stored in a persistent ChromaDB vector database so future documents can be added.

When a user asks a question, the system retrieves the most relevant chunks, sends those chunks to the language model, and generates an answer. The interface displays source file names, page numbers, chunk text, retrieval scores, response time, and retrieval history.

The evaluation layer measures answer relevance, context relevance, faithfulness, hallucination risk, retrieval quality, and response time. These metrics are saved to local history and displayed in a dashboard so engineers can monitor RAG quality over time.